The Refined Gaps

What the Tree of Reality's open boxes become if Chain Theory's candidates hold — a validation of the posture, and a sharper set of predictions

Version: v2, 28 October 2026. **DOI:** 10.5281/zenodo.23724193 · https://zenodo.org/records/23724193 (v1, 25 October 2026, remains at 10.5281/zenodo.23637019; concept DOI 10.5281/zenodo.23637018). **The v1 text below is unchanged.** An addendum is appended at the end. It scores the tests run since v1 against §7's pre-registered table and §8's kill-conditions, records one correction of the Tree's own that those tests found, and pre-registers what has become sharper for v3. No correction from the Chain Theory side has been received as of this version, so v1's descriptions of Chain Theory stand as written; the invitation to correct them carries forward to v3 and is recorded in `docs/chain-theory/07_reading_log.md`. Where v1 says below that it proves nothing new, that describes v1: the addendum reports new Lean theorems (`lean_verify/chain_theory/`) and new public computations (`scripts/chain_theory/`). **Author:** Mark E. Mala (pen name of Ekram Alam), with Claude **Series:** the first successor to *The Staked Gaps* (DOI 10.5281/zenodo.23338726, 13 October 2026), recording the first independent test of its predictions; companion to *The Tree of Reality* (DOI 10.5281/zenodo.23011897, 28 September 2026) **Verification:** this paper proves nothing new and changes no grade. Every claim about the Tree cites a link code (L1–L24), a gap code (G1–G6) or a Lean file in `wonderben-code/convergence-codex`. Every claim about Chain Theory cites a box or theorem number in its public documents of 23 October 2026 and carries Chain Theory's own grade. The one new computation (§4.6) is in `scripts/chain_theory/q3_alpha_count_in_M16.py`, public and dated.

Abstract

A theory of everything was built by one person and an AI and published, machine-verified, on 28 September 2026: a single minimal algebra, forced by self-reference, growing by one repeated operation into the structures of physics. It was published *with its gaps* — twenty-four links, each graded, each unproven residue named — and on 13 October 2026 those gaps were published a second time as pre-registered predictions: where each sits, what shape its filler must take, who would fill it (*"advancing AI, human–machine teams, or the authors"*), roughly when, and what would refute the theory at each site. Ten days later, on 23 October 2026, A. D. Wissner-Gross released *Chain Theory*: an independent, Lean-checked, graded programme deriving Standard-Model structure from finitely many entangled finite-dimensional systems, developed as a human–AI collaboration. Its objects are built from exactly the three dimensions the Tree forces — 2, 4 and 3 — and its results land on exactly the residues the Tree named: three generations, confinement, the Born rule, Lorentzian signature, the three lineages, chirality. The Tree's pre-registered trichotomy for any newly found structure — *extends the walked branch, realises a*

sibling, or fits nowhere — returned its first verdict: *extends the walked branch*. This paper does three things, in the posture the series committed to. **First**, it states precisely what that fit validates — the method: the forced objects are the natural ones, the residues were correctly located, the meta-prediction's agent and timetable were right in kind — and precisely what it does not: no physical identification on either side, none of the six headline gaps G1–G6, no grade, and not the fine-structure constant. **Second**, it takes each candidate *as if correct* and shows that the corresponding gap does not vanish but *sharpens*: a part the candidate would cover, and a residual that is smaller, better specified, and more falsifiable than before. **Third**, it pre-registers those residuals as the next predictions — attribute lists, kill-conditions, and the computation or proof that tests each — so that running the test *is* testing the prediction, in either direction. The result is a theory whose gaps are more precisely known than they were ten days ago, because an independent programme put candidates in them, and whose next edition is earned, not asserted.

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1. The theory, stated

The Tree of Reality proposes that the structures of physics are not axioms but descendants. Where the unification tradition seeks one framework to *contain* the known structures, the Tree derives them from one *ancestor* — a theory of everything in the sense that evolutionary biology is a theory of every species: unification by common descent.

The root is Lawvere's fixed-point theorem: a structure that can refer to itself cannot be empty of fixed points, so self-reference forces existence (L1). The seed is $M_2(\mathbb{C})$, the observable algebra of one qubit — the unique minimal non-commutative algebra over $\mathbb{C}$ (L3). The operation is *take the algebra of endomorphisms*: End(M_b) ≅ M_{b²}, so the seed grows $M_2 \to M_4 \to M_{16} \to M_{256} \to \dots$ (L4), irreversibly (L5). Three lineages run up the tree — what things *are* (End), how they *transform* (Aut), how they are *known* (the inner product) (L7). On the branches: spacetime dimension four, because $Cl_4(\mathbb{C}) \cong M_4(\mathbb{C})$ and no other Clifford algebra is a level of the cascade (L8); Lorentzian signature (L9); the gauge algebra and Pati–Salam structure on $M_{16} = M_4 \otimes M_4$ (L10, L11), with colour as the split $4 \to 3 \oplus 1$ (L13); chirality, three generations, the Higgs sector, anomaly cancellation, the Weinberg ratio 3/8 (L12, L14–L17); the spectral triple and spectral action (L18–L20); the quantum apparatus by GNS, Stone and Schrödinger (L21); and at the frontier, the Einstein equations from the spectral action (L22) and Yang–Mills with its mass gap (L23). Every link is graded against machine-checked Lean 4 proofs in Mathlib.

2. The shape we mapped, and the gaps we left

2.1 The scoreboard, 28 September 2026

Link	Grade	Headline	The named residue (what is *not* proved)
L1	GENUINE	existence from self-reference (Lawvere)	—
L2	PARTIAL	the reflexive domain $D \simeq (D \to D)$	existence as a non-trivial ordered structure beyond the finite levels
L3	GENUINE	the seed $M_2(\mathbb{C})$	—
L4	GENUINE	the cascade $M_2 \to M_4 \to M_{16} \to \dots$	—
L5	GENUINE	irreversibility	—
L6	PARTIAL	why the fourth level (the CCM algebra)	minimality of 4 proved under a stated criterion; the criterion not forced
L7	**POSTULATE**	three lineages: End · Aut · $\langle\cdot,\cdot\rangle$	that these are *the only* lineages
L8	GENUINE	spacetime dimension 4: $Cl_4(\mathbb{C}) \cong M_4(\mathbb{C})$	—
L9	PARTIAL	Lorentzian signature (1,3)	the step excluding the Euclidean alternative (a fork)
L10	PARTIAL	the gauge algebra, embedded	—
L11	PARTIAL	Pati–Salam at M_{16}	the factorisation 4·4 chosen among $a \cdot b \cdot c = 16$
L12	**POSTULATE**	chirality — which factor is "left"	the choice of factor
L13	GENUINE	colour: the 4 splits as $3 \oplus 1$	—
L14	**POSTULATE**	three generations — the imaginary quaternions	the count itself
L15	PARTIAL	the Higgs sector and coupling window	any scalar potential; the vacua
L16	PARTIAL	anomaly cancellation	—
L17	PARTIAL	the Weinberg angle 3/8 as a trace ratio	the coupling matching at unification
L18	PARTIAL	the spectral triple	—
L19	PARTIAL	the spectral action: the exponential forced	forced for one algebra at one generation only
L20	PARTIAL	the Boltzmann measure of the action	—
L21	PARTIAL	Hilbert space, GNS, Stone, Schrödinger	*the Born rule, Gleason and Wigner are cited physics, not theorems*
L22	◇ **OPEN**	the Einstein equations from the spectral action	gap G5, gated on G2
L23	◇ **OPEN**	Yang–Mills and the mass gap	walls W1–W4; gaps G1, G3, G4

Link	Grade	Headline	The named residue (what is *not* proved)
L24	PARTIAL	the numbers	no coupling constant computed

6 GENUINE · 13 PARTIAL · 3 POSTULATE · 2 OPEN.

2.2 Three kinds of gap, three kinds of prediction

The Staked Gaps (13 October) distinguished them:

- **The six headline gaps G1–G6** — analytic and representation-theoretic obstacles in the proof chain (the infinite-volume limit; the unbounded-operator library; a comparison bound of new shape; a uniform sub-top spectral estimate; the heat-kernel expansion of the spectral action; the isotypic bimodule and its K-theory). Each was staked with an *attribute list*, a *kind* (pinned / bounded / open-directional), and a *kill-condition*.
- **The residues** — "Every PARTIAL link of the capstone's §4 carries a named, machine-checkable remainder, and the prediction covers them too: as the boxes fill and the residues close, the wood of the tree hardens."
- **The sibling branches** — alternative descendants of the seed, proved to exist as mathematics at four forks, predicted to be "candidates for physics not yet found."

2.3 Who, and when — the meta-prediction

> *"As automated mathematical reasoning continues its current trajectory, the gaps named here will be closed — in the shapes predicted — soon (on the order of years, not decades) … **The agent of closure will be advancing AI, human–machine teams, or the authors.**" (The Staked Gaps §7)*

And the protocol for a fill:

> *"Anyone — human, machine, or the two together — who closes a gap in the shape predicted … confirms a pre-registered prediction; the dated, Bitcoin-anchored record of this paper establishes the priority of the prediction, **not a claim on the solver's work**." (§8)*

2.4 Where anything new must land — the trichotomy

> *"Physics not yet known will, when found, do one of three things. **Either it extends the walked branch** — a new structure of our own lineage, joining the tree where the tree already has a place for it, strengthening it; **or it realises a sibling** … ; **or it fits***

> ***nowhere on the tree at all*** *… The first two confirm the theory, and are predicted. The third refutes the generative claim itself." (The Tree of Reality §9)*

With the boundary that keeps it honest: "the tree is refutable on the branch it walks, and never excused by the branches it merely permits."

That is the shape we mapped. The posture was stated as a posture — "the analogy claims a *posture*, not a *stature*" — and what follows is the first occasion on which the posture was tested by something we did not build.

3. What arrived, ten days later

On 23 October 2026 A. D. Wissner-Gross released *Chain Theory*: a 24-page preprint (*A Parameter-Free Candidate for the Fine-Structure Constant from Entangled Qudits*), a 754-page working-draft compendium (release 0.1; 100 theorems, 64 tied to Lean declarations; 109 conditional theorems; 198 open results), an introductory slide deck, and a self-contained Core Lean 4 file checking the preprint's central count. The announcement followed on 24 October.

Its axiom: *"Nothing exists but finitely many maximally entangled pairs of qudits, some qudits shared, known only by distilling them."* From this and six definitions it obtains a *read cap* (a read fixes both ends of a pair only when $d_1 d_2 \le 6$, by Størmer–Woronowicz–Choi), six *genres* of link labelled by end dimensions drawn from {2, 3, 4} (Conjecture 0, box 1.7), the group $S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$ as the symmetry of the qutrit–qubit pair (theorems 64.2, 65.1), and — as conditional theorems — the Born rule, the Lorentz group and signature, three generations as a bound, confinement, curvature and a cosmological term. Its headline is an exact number, $(79 + 48\sqrt{2} + 42\sqrt{3} + 3\sqrt{6})/31104$, a theorem of the definitions, *conjectured* to be the fine-structure constant at zero momentum transfer (0.60σ from CODATA 2022). Its acknowledgments state that the work "was developed as part of a long-term human-superintelligence collaboration supervised by the human author."

Neither programme cites the other before this paper. Chain Theory names "the spectral action on a finite noncommutative geometry" as a sibling tradition; the Tree grows from it.

4. The fit

We now hold Chain Theory against the shape of §2, object by object, and record where it fits, where it does not, and which fits are generic.

4.1 The objects: built from exactly what the Tree forces

Chain Theory's Conjecture 0 is that link ends have dimension 2, 3 or 4 — found from its move table, graded a conditional theorem, with no generative reason given for the set. The Tree's cascade produces levels of dimension 2, 4, 16, … (L4, GENUINE); colour splits the 4 as 3⊕1 (L13, GENUINE). So **the Tree generates exactly {2, 4} as levels and {3} as a split of a level, and nothing of dimension 5, 6, 7 or 8 below 16.** An independent programme, with no generative principle, found by exhaustion the same three dimensions the Tree forces by construction. Chain Theory's ququart is End(M_2); its qubit is the seed; its qutrit is the colour triplet.

4.2 The residues: candidates land where the residues were named

Tree residue (28 Sep)	Chain Theory candidate (23 Oct)	Its grade
L14 · "three generations … a physical postulate; the mathematics proves what follows from it" — *the count itself*	the generation bound at the hub dimension: a ququart hub assembly admits at most the third generation (197.3); three at the core triangle (83.1); three *as* the count held open (197.5)	197.3 theorem; 83.1 theorem on the roster; 197.5 open
L23 · a wall, no mechanism	confinement as dark-genre readability: the (3,3) link has $d_1 d_2 = 9 > 6$, so no single read fixes both ends (171.1)	conditional
L21 · "the Born rule, Gleason and Wigner are cited physics, not theorems here"	the Born readout from the wash (35.1, 40.1), premises named: the Hilbert space of a qudit and the pair as a ray	conditional
L9 · the fork: Euclidean not excluded	the Lorentz group as isometries of the Hermitian split (54.1); one against three at every step (55.1); the Pell-lattice argument that only the indefinite form has infinitely many integer isometries	54.1 conditional; 55.1 theorem
L7 · "that End, Aut and ⟨·,·⟩ are *the only* lineages"	Definition 1.6, "known": what all reads fix jointly under every matching and exchange — pairs / matchings / reads as the three primitives	definition
L12 · which factor is "left"	parity as the improper element (59.1); three involutions with CPT their product (61.1)	59.1 conditional; 61.1 theorem

Six of the Tree's nine non-GENUINE-non-headline openings received a candidate aimed at the named residue. None of the candidates was aimed at the Tree; each is a result of Chain Theory about its own networks. That they land where they do is the content.

4.3 The trichotomy: *extends the walked branch*

The capstone's first outcome reads: "a new structure of our own lineage, joining the tree where the tree already has a place for it, strengthening it." Chain Theory's objects — qubit, pair of qubits, ququart, qutrit, $S(U(3)\times U(2))$, $SU(4) = Spin(6)$ — each have a place on the walked branch (L3, L8, L4, L13, L10–L11, L11). None realises an unwalked sibling (no quaternionic seed, no Riemannian signature, no alternative cut of M_{16}). None fits nowhere. **The first independent**

test of the trichotomy returned the first confirming outcome. The boundary of §2.4 is respected: nothing here is a failure on the walked branch reclassified as a sibling.

4.4 The meta-prediction: agent and timetable, right in kind

The staked agent of closure was "advancing AI, human–machine teams, or the authors." Chain Theory describes itself as a human–superintelligence collaboration and checks its computations in Lean. The staked timetable was "soon (on the order of years, not decades)." Candidates for six residues arrived ten days after the predictions were staked and twenty-five days after the tree was published. We record this as the meta-prediction materialising *in kind* — the kind of agent, the kind of method, the direction of the timetable — and we are explicit that it materialised on *residues*, not on the six headline gaps, whose timetable is unaffected (§4.5).

4.5 What did not fit

The six headline gaps G1–G6 received no candidates. Chain Theory's confinement mechanism is of a different kind from the continuum-limit and spectral-gap estimates G1, G3 and G4 ask for. Its gravity (depth metric, graph-Laplacian curvature, cosmological term at the depth exponent) is a route disjoint from the heat-kernel expansion G5 predicts. It uses neither unbounded operators (G2) nor the spectral triple (G6). The headline predictions stand exactly as staked.

The number did not transfer (§4.6). **No physical identification was confirmed** on either side (§5).

Generic and non-generic fits. The Standard-Model group as S(U(3)×U(2)) is a generic consequence of having a 3 and a 2 — the stabiliser of a 3+2 split in SU(5) (Baez–Huerta 2010, which Chain Theory cites) — and any finite programme with a qutrit and a qubit reaches it. That fit is weak evidence. Three fits are not generic: (i) the pair of qubits as *four*-dimensional spacetime — the Tree's theorem about Clifford algebras, Chain Theory's spin-c group of the pair, with nothing in "two qubits" forcing "four dimensions" in general; (ii) the register set *exactly* {2, 3, 4} — found by exhaustion on one side, generated on the other, with neither having a reason to stop at 4 except that it does; (iii) the threshold 6 marking the visible world on both sides — Chain Theory's read cap from positive-map decomposability, the Tree's $3 \otimes 2$ block from the colour split, with no shared premise. A fourth non-generic fit emerged from the computation in §4.6.

4.6 One computation: the α object inside the Tree

We rebuilt Chain Theory's 172-record count from its own rules — letters at four slots (8, 3, 8, 3), 169 products and 3 commutators, exchanges X and Y as 172 × 172 commuting involutions — and

reproduced it exactly: sectors 79, 48, 42, 3; $1/\alpha = 137.035\ 999\ 189\ 566\ 155\ldots$ We then realised the object inside the Tree's $M_{16} = (3 \oplus 1) \otimes 2 \otimes 2$ by explicit matrices. Three facts:

1. The qutrit and both read qubits are present with the right letter algebras and commutation structure, so **the numerator transfers unchanged** — it depends on the end dimensions alone.
2. The denominator's unread third link needs a third qubit entangled with the qutrit. **M_{16} has two; the third first appears at M_{256} — the Tree's fourth level, the level at which L6 independently places the Chamseddine–Connes–Marcolli finite algebra.** Neither programme had reason to expect the other's object to complete there. This is the fourth non-generic fit.
3. The 169 products, realised as matrices, span only 66 dimensions: Chain Theory's two *copies* of the one shared qutrit — "one copy per link" (Definition 1.3) — collapse to a single M_3 factor. **The number lives in Chain Theory's record space, defined by "shared" (1.3) and "known" (1.6), not in the operator algebra of the shared system.** The Tree supplies the object; it does not supply the number. For Chain Theory this localises exactly which clauses a derivation of α rests on. For the Tree it sharpens L24 (§6.4).

5. What this validates, and what it does not

We owe the reader the boundary before the claims.

Not validated. No physical identification, on either side: that M_4 is spacetime, that the 3 is colour, that the (3,2) link is the photon, that the number is α — all remain conjectures, graded as such by both programmes. Not one of G1–G6. Not one grade on the Tree: the scoreboard is $6 \cdot 13 \cdot 3 \cdot 2$ today as it was on 28 September. Not the fine-structure constant: the Tree does not derive it, and §4.6 shows why it cannot from the algebra alone.

Validated — three claims about the method, each bounded.

1. **The forced objects are the natural ones.** The Tree's distinctive claim is generative: that 2, 4 and 3 are not chosen but produced. An independent programme with no generative principle, searching its own move table, found exactly those three dimensions and no others. If the Tree had forced {2, 4, 16} with colour as $16 \to 15 \oplus 1$, or if Chain Theory had needed a register of 5, the two would have missed. They did not. This is the kind of evidence a generative claim can receive short of proof: the things it says must exist are the things someone else found.
2. **The residues were correctly located.** Six named residues received candidates aimed at exactly what the residue said was missing — the count of generations, a mechanism for confinement, a derivation of the probability rule, an exclusion of the Euclidean signature, a reason for three lineages, a reason for "left". The Tree did not say "something is missing

here"; it said *what* was missing, and candidates for *that* arrived. This validates the grading discipline: a PARTIAL row that names its residue precisely is a row someone else can fill.

3. **The meta-prediction was right in kind.** Agent (human–AI collaboration with formal checking), method (Lean), direction (candidates within weeks of staking). It was not tested on G1–G6, where the timetable remains open, and we do not claim it was.

The validation is of the *posture*, not the *stature* — exactly the distinction *The Staked Gaps* drew. A theory that specifies its own missing pieces precisely enough that an independent programme's results can be matched to them, attribute by attribute, is a theory whose gaps are doing work. That is now demonstrated once. Whether the pieces *fit* — whether the candidates hold and the residuals close — is what §6 and §7 put at risk.

6. The refined gaps

The Mendeleev move is not "a candidate arrived, box filled." It is: *a candidate of this shape lets us predict the box's finer properties*. Conditional on each candidate being correct, the corresponding gap decomposes into the part the candidate would cover and a residual — smaller, sharper, and killable. We state each residual in the format of *The Staked Gaps*: prediction kind, attributes, kill-condition, and the test (from `docs/chain-theory/06_questions_to_work_on_together.md`) whose outcome decides it.

6.1 L14 · three generations

As staked: POSTULATE — the count is not derived; the candidate reason is that $\mathbb{H} \subset M_2(\mathbb{C})$ has three imaginary units. **If the candidate holds (197.3):** the count is bounded above by three, from the hub dimension four — i.e. from $M_4 = M_2 \otimes M_2$, the Tree's first cascade step. **The gap becomes two:** - **(a) the bound, transferred.** *Pinned.* Attributes: a Mathlib theorem on tensor factorisations of $M_4 = M_2 \otimes M_2$ stating that at most three independent "attachments" of the kind Chain Theory's hub assembly makes are admitted. Kill-condition: the bound depends essentially on Chain Theory's network moves and has no statement about M_4 alone. Test: Q4. - **(b) exactly three.** *Pinned.* Attributes: the three attach sites of Chain Theory's closed core triangle (83.1, 197.4), realised as operators, satisfy the quaternion relations $i^2 = j^2 = k^2 = ijk = -1$ up to the twirl — so that Chain Theory's open result 197.5 and the Tree's postulate are one statement. Kill-condition: the three attach-site operators commute, or generate something other than $\mathbb{H}$. Test: Q5. **What moves if both hold:** L14 → GENUINE. If (a) holds and (b) fails: L14 → "bounded above by theorem; selection open" — a real grade change, and the postulate is replaced by a smaller one.

6.2 L23 · Yang–Mills and the mass gap

As staked: ◇ OPEN — a wall (W1–W4) with gaps G1, G3, G4 behind it; no mechanism. **If the candidate holds (171.1):** confinement has a *reason* — on the colour block $M_3 \otimes M_3$ (the Tree's qutrit twice), $d_1 d_2 = 9$ exceeds the read cap, so no single act of knowing fixes both colour ends. **The gap becomes two:** - **(a) the mechanism.** *Bounded.* Attributes: a statement about the spectral triple (A, H, D) restricted to the colour block — our ranked candidates: (i) no conditional expectation onto a mutually-unbiased-basis sub-algebra of $M_3 \otimes M_3$ commutes with D on that block; (ii) the positive-map decomposability threshold reappears as a property of the bimodule structure (L18). Kill-condition: the read cap has no statement in the algebraic picture — it is a feature of the network of copies that the operator algebra does not see. Test: Q6. - **(b) the continuum.** *Unchanged.* G1, G3, G4 exactly as staked. The candidate speaks to *why* there is a gap, not to *how large* it is or whether the infinite-volume limit exists. We predict that a successful (a) does not shorten (b)'s timetable. **What moves if (a) holds:** L23 → "mechanism stated, continuum limit open" — from a wall to a staircase with its first step built.

6.3 L21 · the Born rule

As staked: PARTIAL — Hilbert space by GNS from the inner product; the Born rule, Gleason and Wigner *cited*. **If the candidate holds (35.1, 40.1):** the probability rule is an output of the wash, given the Hilbert space of a qudit and the pair as a ray. **The gap becomes one, and small:** *Pinned.* Attributes: the chain *GNS (Tree) → Hilbert space of a qudit → wash (Chain Theory) → Born readout* closes with nothing used twice and nothing cited — the Tree supplies exactly the import Chain Theory names, and Chain Theory supplies exactly the rule the Tree cites. Kill-condition: a second import in Chain Theory's premises ("the pair as a ray," or the twirl's invariance) that the Tree's inner-product lineage does not already provide. Test: Q8. **What moves if it holds:** L21 → GENUINE for the probability rule; the assumptions ledger loses entry 59.

6.4 L24 · the numbers

As staked: PARTIAL — two numbers computed (the Weinberg ratio, a Higgs window); no coupling constant. **What §4.6 established:** the α object's ingredients are forced by the cascade (qutrit, two qubits at M_{16}, the third at M_{256}), but the number lives in Chain Theory's record space, not in the algebra. **The gap becomes precise:** *Open-directional.* Attributes: a construction, over M_{256}, of a record space in which the one shared qutrit appears as *one copy per link* — the content of Chain Theory's Definition 1.3 — from the Tree's third lineage (the inner product) and nothing else. If it exists, the Tree derives the 172-record space and with it the numerator and denominator; if it does not, the Tree has identified "one copy per link" as the one extra-algebraic clause in the derivation of α. Kill-condition for the *Tree's* side: the inner product lineage provably yields only the operator algebra of the shared system (rank 66 of the 169 products) and no copy structure. Test: a follow-on to Q3, stated in `docs/chain-theory/`

`08_results_first_pass.md`. **What moves:** nothing until the construction exists. But the gap has changed character: from "no coupling constant" to "one named clause stands between the Tree and a coupling constant."

6.5 L9 · Lorentzian signature

As staked: PARTIAL — the fork; Euclidean not excluded. **If the candidate holds (54.1, ch. 54):** the indefinite form on the lattice of counts admits infinitely many integer isometries (a Pell family) while the definite form admits finitely many, so the signature "reads Lorentzian by existence." **The gap becomes one:** *Pinned*. Attributes: the Tree's Clifford lattice in M_4 (the integer structure of $\gamma_1 \ldots \gamma_4$ in `F4_1e_CliffordMatrix.lean`) carries an indefinite integer form with an infinite isometry group and a definite one with a finite group. Kill-condition: both forms on that lattice have infinite isometry groups, or both finite. Test: a lattice computation on the γ basis — straightforward. **What moves if it holds:** L9 → GENUINE, by a route the Tree had not tried.

6.6 L7 · the three lineages

As staked: POSTULATE — that End, Aut and $\langle \cdot, \cdot \rangle$ are *the only* lineages. **If the candidate holds (Definition 1.6):** a structure is a thing (pairs), acted on (matchings), and known only through what the action leaves fixed (reads/"known") — three primitives, no fourth. **The gap becomes one:** *Pinned*. Attributes: a theorem that an object equipped with an action and a knowledge clause of Chain Theory's form has exactly three aspects — carrier, symmetry, invariant pairing — and that any fourth is a function of these. Kill-condition: a fourth primitive that is not derived. Test: Q9. **What moves if it holds:** L7 → GENUINE; the Tree's only *structural* postulate becomes a theorem with a knowledge clause as its premise.

6.7 L12 · chirality

As staked: POSTULATE — which factor of $M_4 \otimes M_4$ is "left" is chosen. **If the candidate holds (59.1, 61.1):** parity is the improper element of the Hermitian split's isometry group; CPT is the product of three involutions. **The gap becomes one:** *Bounded*. Attributes: the improper element, realised on $M_4 \otimes M_4$, either (i) exchanges the two factors — in which case "left" is a convention, as *The Forced and the Chosen* already argues — or (ii) acts non-trivially within one factor, singling it out. Kill-condition: the improper element has no action on the factorisation at all. Test: a computation on the γ basis.

6.8 The fourth level — a new prediction from §4.6

Not previously staked. The Tree's L6 places the Chamseddine–Connes–Marcolli finite algebra — the full Standard-Model fermion content as a bimodule — at the fourth level M_{256}, under a stated minimality criterion. §4.6 found that Chain Theory's α object, complete with its third qubit, first fits at the same level. **Prediction:** *Pinned*. Attributes: Chain Theory's sixteen

particle webs (17 particles), each a network of links with ends in {2, 3, 4}, embed in $M_{256} = M_{16} \otimes M_{16}$ with the hub assemblies as M_4 factors and the core triangles in the colour blocks — and the embedding respects the CCM bimodule's isotypic decomposition (G6). Kill-condition: a particle web of Chain Theory requires more tensor factors than M_{256} provides, or fits only at M_{65536}. Test: a dimension count per web, then the embedding. This is the first prediction the two programmes generate *jointly*: neither could have stated it alone.

6.9 In the other direction: what the Tree predicts for Chain Theory's open items

The refinement runs both ways. Taking the Tree's theorems as correct:

- **Conjecture 0 (1.7).** *Prediction:* no web of Chain Theory will ever require an end of dimension 5, 6, 7 or 8; the next admissible register after 4 is 16. *Kill-condition:* a web that needs a 5-dimensional end. Test: Q10, a census of the move table.
- **Conjecture 2 (1.9).** *Prediction:* the four generators of the (2,2) register (ch. 3-5) are Clifford generators of $Cl_4(\mathbb{C})$ on $M_2 \otimes M_2$, and no (d,d) register other than (2,2) carries a Clifford structure that is a cascade level. Already checked in part: the compendium's β is the Tree's γ_1 entry for entry (Q2). *Kill-condition:* a Clifford structure at (3,3) or (4,4) that is a cascade level.
- **The Weinberg angle (not derived in the compendium).** *Prediction:* if Chain Theory's W and Z "offices" (164.5: ratio exactly ½) are the traces $Tr(T_3L^2)$ and $Tr(Q^2)$ on its particle webs, their ratio at the unification reading is 3/8 and at the hypercharge reading 3/5. *Kill-condition:* a different ratio from the same offices.
- **The step below the axiom.** *Prediction:* Chain Theory's Definition 1.6 is satisfiable only on a carrier admitting a reflexive structure $D \simeq (D \to D)$, so that Lawvere's theorem applies and the existence of at least one qudit is forced rather than posited. *Kill-condition:* a model of the six definitions on a carrier with no fixed-point property.

7. The predictions, pre-registered

#	Residual	Kind	Test	Confirms	Refutes (at that site)
R1	L14(a) the ≤3 bound on M_4	pinned	Q4	bound is a theorem of $M_2 \otimes M_2$	bound needs the network
R2	L14(b) attach sites = i, j, k	pinned	Q5	quaternion relations hold up to twirl	sites commute / not $\mathbb{H}$
R3	L23(a) read cap on the spectral triple	bounded (2 shapes)	Q6	a statement about (A, H, D) on the colour block	no algebraic statement
R4	L23(b) continuum unchanged	—	—	G1/G3/G4 timetable unaffected	(a) shortens (b) — would be a surprise, not a refutation

#	Residual	Kind	Test	Confirms	Refutes (at that site)
R5	L21 the Born circle	pinned	Q8	GNS → wash closes with no import	a second hidden import
R6	L24 record space over M_{256}	open-directional	Q3′	"one copy per link" from the inner product	only the operator algebra (rank 66)
R7	L9 Pell family on the γ lattice	pinned	lattice computation	indefinite infinite, definite finite	both infinite or both finite
R8	L7 three aspects from a knowledge clause	pinned	Q9	carrier / symmetry / pairing, no fourth	a fourth primitive
R9	L12 the improper element on the factorisation	bounded (2 shapes)	γ computation	exchanges factors, or singles one out	no action
R10	Fourth level: particle webs in M_{256}	pinned	dimension count + embedding	all 16 webs fit, respecting G6's decomposition	a web needs M_{65536}
R11	Chain Theory: no register 5–8	pinned	Q10	census confirms	a 5-dim end
R12	Chain Theory: Weinberg 3/8 from the offices	pinned	trace identification	3/8 and 3/5	other ratios

The gated order, as before. R7, R9 and R10's dimension count are computations and come first. R1, R3, R5, R8 are Mathlib proofs. R2, R6, R10's embedding, R11, R12 need both programmes' objects side by side and are the joint work.

7.1 First results, 25 October 2026

Two residuals were pure computations and were run on the day of writing (`scripts/chain_theory/r7_r9_lattice_and_parity.py`, public). Both came back as *refinements* — the outcome §6 said a correct candidate should produce.

R7 — holds as stated, and sharpens. On the Tree's Clifford lattice, the Euclidean form (4,0) has exactly 384 integer isometries (finite, by enumeration); the Lorentzian form (1,3) and the split form (2,2) each admit an integer isometry of infinite order (spectral radii 5.83 and 5.16, explicit matrices in the script). So Chain Theory's existence argument *excludes the Euclidean branch* at the fork L9 — a theorem-grade exclusion the Tree did not have — and does *not* choose between the indefinite signatures (1,3) and (2,2), which share $Cl_4(\mathbb{C})$. L9's residue narrows to "select among the indefinite three"; it does not close.

R9 — answered, and re-posed. With the Tree's γ-basis, the improper element $P = \gamma_1 = \sigma_3 \otimes I_2$ acts within the first tensor factor of $M_4 = M_2 \otimes M_2$ (shape ii) and anticommutes with $\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4 = \sigma_2 \otimes \sigma_2$. But γ_5 is not aligned with the factorisation — the Weyl split is not a tensor-factor split — and which factor P lives in is a property of the basis. So "which factor is left" is not

a question M_4 can answer, and R9 was posed on the wrong object. It is re-posed at $M_{16} = M_4 \otimes M_4$, where L12's choice actually lives, as **R9′**: *does the improper element of Cl_4, lifted to M_{16}, exchange the two M_4 factors or single one out?* We record this as a correction to this paper found by its own test, on the day of writing, and leave the original R9 in the table above so the correction is visible.

8. What kills this paper

Three layers, stated in advance.

If the candidates fail on their own side — if Chain Theory's 197.3, 171.1, 35.1 or 54.1 do not survive its own further work — then §6's conditionals are vacuous for those rows and the gaps revert to their 13 October statement. §4's validation of the *method* survives, because it concerned where candidates were aimed, not whether they hold; §5's three claims survive for the same reason. We would record the reversion under this paper's DOI lineage.

If the candidates hold but a residual fails — if the ≤3 bound is a theorem of Chain Theory's networks but has no statement on M_4 (R1 refuted); if the read cap has no algebraic shadow (R3 refuted); if the attach sites are not quaternionic (R2 refuted) — then the Tree is refuted *at that site*, on the branch it walks, and may not be excused: the capstone's boundary applies in full. Each such refutation would be recorded with the same prominence as a confirmation, and the link's grade would stand or fall accordingly.

If the candidates hold and the residuals hold — then grades move, and a second edition of *The Tree of Reality* is earned. We commit to publishing it only then.

And the trichotomy remains live: the next independent programme to appear is held to the same three outcomes, and a structure that fits nowhere refutes the generative claim regardless of anything in this paper.

9. Caveats, both programmes

Chain Theory's α number is a theorem of its definitions and its identification with α(0) a conjecture, as its author states; nearby wordings of the definitions give 1/α between 135.89 and 146.52, each excluded by a clause and not by comparison with experiment; four of six sector identifications are "by name"; many compendium theorems are, in its own words, "a census of a finite catalog, not a proof for every web." The Tree computes no coupling constant; thirteen of its links are PARTIAL, three POSTULATE, two OPEN; its gravity is gated on an unbuilt library; its identification with physics is exactly as conjectural as Chain Theory's. Nothing in this paper has been verified beyond the two programmes' own Lean files and the one public computation of §4.6, which check finite mathematics and not physical identification. The ten days are a fact

about timing. The fit is a fact about shape. Neither is a fact about truth; §7 is where truth is put at risk.

10. Provenance

This paper is deposited on Zenodo under its own DOI as a successor to *The Staked Gaps*, and committed to `wonderben-code/convergence-codex`, where every push is anchored to the Bitcoin blockchain by OpenTimestamps; the commit precedes the deposit. It is published in the open, as every paper of this series is, and was sent to the author of Chain Theory on publication with an invitation to correct any misstatement of his work; corrections received will be incorporated in v2 under the same concept DOI and recorded in `docs/chain-theory/07_reading_log.md`. The dated record establishes the priority of the *predictions*; it makes no claim on anyone's solutions, and every result of Chain Theory cited here is his.

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Addendum (version 2, 28 October 2026): The Residuals, Tested

Seven tests run three days after v1: what held, what sharpened, what failed, and one error of our own

Summary of the addendum

Version 1 pre-registered twelve residuals (R1–R12, §7), each with the result that would confirm it and the result that would refute the Tree at that site (§8). It promised that a refutation would be recorded with the same prominence as a confirmation. Between 25 and 28 October 2026 the residuals that could be run quickly were run: as Lean theorems where they were proofs, and as public scripts where they were computations. One further test was run that v1 had not listed, and it found an error in the Tree itself.

The outcomes, in one line each:

- **R1** (the three-generation bound on M_4): **holds, in a re-posed form.** The counting is a theorem about frames of M_4 and M_3, machine-checked. The tensor-factor form v1 asked for is provably impossible.
- **R2** (the attach sites are i, j, k): **refuted.** No operators on Chain Theory's three-dimensional generation space can satisfy the quaternion relations, and this is proved in Lean. Per §8 this is a refutation at the L14 site, conditional on Chain Theory's own account of generations, which is open on its side.
- **R3** (the read cap on the colour block): **confirmed in shape (ii), refuted in shape (i).** The decomposability threshold reappears across the Tree's bimodule cut exactly at the colour block; the statement about the Dirac operator is false.
- **R9 → R9′** (the improper element at M_{16}): **answered.** It singles out one factor and does not exchange them. Exchanging the factors is the work of a different class of operations, so L12 is re-posed as two separate choices.

- **R11** (no register 5–8): **consistent from the Tree's side.** As a by-product, Chain Theory's Conjecture 0 narrows the Tree's L11 from four alternatives to two.
- **R4** (the continuum is unchanged): **consistent.** Nothing here touches G1, G3 or G4.
- **R5, R6, R8, R10, R12: not yet run.** They are carried to v3 (§A8).
- **Not pre-registered: the Higgs window (L15).** Read physically, with Standard-Model running, the Tree's window **excludes the measured Higgs mass**. This is a correction of the Tree, found by our own test. The same section compares Chain Theory's Higgs quartic, computed from its own numbers, with the window.

No grade moves. The scoreboard stays at 6 GENUINE · 13 PARTIAL · 3 POSTULATE · 2 OPEN. Section A8 lists what has become narrower or newly computable, as the work to attempt before v3.

All files are public in `wonderben-code/convergence-codex`:

- Lean: `lean_verify/chain_theory/Q4HubBound.lean` and `Q5TwoThrees.lean`, built in the estate as the `ChainTheory` library; every theorem depends only on `propext`, `Classical.choice` and `Quot.sound`.
- Scripts: `scripts/chain_theory/` (`r9prime_parity_at_M16.py`, `q6_read_cap_on_colour_block.py`, `q1_quartic_vs_window.py`, `q5_two_threes.py`, `q10_registers_vs_cascade.py`).
- Write-up: `docs/chain-theory/09_results_second_pass.md`.

A1. The scorecard

#	Residual (v1 §7)	Confirms / refutes, as pre-registered	Outcome
R1	L14(a) the ≤3 bound on M_4	bound a theorem of $M_2 \otimes M_2$ / bound needs the network	**Holds, re-posed.** The M_4 half (the hub is saturated) is a theorem of M_4 alone; the count of three uses the network's triangle; the tensor-factor form is impossible
R2	L14(b) attach sites = i, j, k	quaternion relations hold / sites commute or are not $\mathbb{H}$	**Refuted**, and more strongly than the kill-condition: no quaternion action exists on the generation space
R3	L23(a) the read cap on the spectral triple	a statement on the colour block / no algebraic statement	**Shape (ii) confirmed** (the bimodule cut); **shape (i) refuted** (the Dirac operator)
R4	L23(b) continuum unchanged	G1/G3/G4 unaffected	**Consistent**
R5	L21 the Born circle	—	not run
R6	L24 record space over M_{256}	—	not run

#	Residual (v1 §7)	Confirms / refutes, as pre-registered	Outcome
R7	L9 Pell family	—	run in v1 (§7.1): holds and sharpens
R8	L7 three aspects	—	not run
R9′	L12 the improper element at M_{16}	exchanges the factors, or singles one out / no action	**Singles one out.** The exchange belongs to anti-automorphisms; L12 is re-posed
R10	fourth level, the webs in M_{256}	—	not run
R11	no register 5–8	the census confirms / a 5-dimensional end	**Consistent** from the Tree's side; by-product: L11 narrows from 4 to 2
R12	Weinberg 3/8 from the offices	—	not run

A2. R1 · the bound transfers, as a theorem about frames

What Chain Theory's bound rests on. Theorem 197.3 reads: *"a hub assembly is a ququart carrying a qubit pair among its qubit legs; the pair takes two of its four ends and every move applies at a site with spare alone, so a hub assembly admits at most the third generation at the hub dimension four."* The counting is Chain Theory's *valence law*: at most d pairs at a qudit of dimension d. The compendium reads this law off Definition 1.5, under which a read gives a qudit its only basis, *"one direction for each pair the qudit meets"*.

Monogamy is respected by the copy model. A shared qudit *"holds one copy of itself for each of its pairs … much as a four-dimensional system, which has room for two qubits, can be maximally entangled with two separate qubits at once."*

So "a ququart has four ends" can be read in two ways:

- **additively**, as four directions of a frame; or
- **multiplicatively**, as tensor factors.

v1 posed R1 in the second reading. The Lean file `Q4HubBound.lean` tests both.

Theorem	Statement
A `card_le_of_orthogonal_idempotents`	nonzero idempotents of $M_n(K)$ with $P_iP_j = 0$ for $i \neq j$ number at most n (the valence law, read additively)
B `hub_saturated`	an in-leg and three legs in M_4 leave no nonzero idempotent orthogonal to all four (no spare at a full hub)
B′ `hub_legs_le_three`	beside an in-leg, at most three legs fit in M_4
C `core_spare_le_one`	in M_3, with two core ends taken, at most one further end fits
C′ `core_attachments_le_three`	three such core qutrits admit at most three attachments in total
`hub_hypotheses_satisfiable`	the four diagonal units of M_4 meet B's hypotheses (B is not vacuous)

Theorem	Statement
D `dvd_of_unital_algHom`	a unital algebra map $M_m(K) \to M_n(K)$ in characteristic 0 exists only if m divides n
D′	there is no unital $M_3(\mathbb{C}) \to M_4(\mathbb{C})$ and no unital $M_8(\mathbb{C}) \to M_4(\mathbb{C})$

Result. The bound transfers as a theorem about **frames**:

- the hub assembly fills M_4 ($1 + 3 = 4$, nothing spare);
- the closed core triangle leaves one direction at each of its three qutrits;
- so there are at most three attachments.

In the tensor reading, R1's attribute fails. Theorem D shows the qubit pair already exhausts $M_4 = M_2 \otimes M_2$, and a qutrit fits inside a ququart only as a *corner*, never as a tensor factor. That corner is the Tree's own L13 split, $4 = 3 \oplus 1$.

Against §8's kill-condition. R1's kill-condition was "the bound needs the network and has no statement about M_4 alone". It is half met:

- M_4 alone carries the saturation of the hub (Theorem B);
- the count of three needs the triangle's shape, which Theorem C′ takes as a hypothesis, from the network.

We record R1 as holding in a re-posed form. The definition that carries the weight is Chain Theory's Definition 1.5, read on the carrier $\mathbb{C}^4$.

A3. R2 · the attach sites are not the quaternions: a refutation, recorded as promised

v1's §8 committed, before any test was run: *"if the attach sites are not quaternionic (R2 refuted) … then the Tree is refuted at that site, on the branch it walks, and may not be excused … Each such refutation would be recorded with the same prominence as a confirmation."* This section does that.

Where each three lives, in Chain Theory's own words. The derivation of Chain Theory's Theorem 2.9 says: *"at a qubit end the commutator phase of two reads is a sign and their product one further traceless operator fixed under every matching, while a qutrit end carries a cube root."*

- **The Tree's three** (L14) is the first structure. The imaginary quaternions i, j, k are a qubit's three traceless directions: two reads that anticommute, and their product.
- **Chain Theory's three** (197.4) is the second. The generations are the three *qutrit* corners of the closed core triangle, spanning *"a generation space of dimension three"*, carried cyclically by the shift (197.2). Their mixing is the order-three Fourier matrix (166.1).

What the Lean file `Q5TwoThrees.lean` proves:

- **(A)** `twist_pow_eq_one`**.** If invertible A and B on an n-dimensional space satisfy AB = c·BA, then $c^n = 1$. So the commutator phase at a d-dimensional end is a d-th root of unity: a sign at a qubit, a cube root at a qutrit. This is a second, independent proof of the clause quoted above.
- **(B)** `no_anticommuting_pair_of_odd`**.** In odd dimension, no two invertible operators anticommute.
- **(C)** `no_quaternion_action_of_odd`**.** There is no unital $\mathbb{R}$-algebra map $\mathbb{H} \to M_n(\mathbb{C})$ for odd n, and in particular none into $M_3(\mathbb{C})$.

The verdict. R2's kill-condition was "the three attach-site operators commute, or generate something other than $\mathbb{H}$". The truth is stronger. Theorem C shows that **no operators whatever on Chain Theory's three-dimensional generation space realise the quaternion relations**, not even up to a phase. The two threes are different structures:

- the Tree's: a qubit's three anticommuting directions, with commutator phase −1;
- Chain Theory's: a qutrit triangle's three sites, with phase e^(2πi/3).

What is refuted, and what is not. R2 predicted that Chain Theory's open result 197.5 ("three generations as attach sites") and the Tree's postulate L14 ("three generations are the imaginary quaternions") are one statement. They are not.

The two programmes therefore give **different accounts of why there are three generations**, and the two accounts cannot be one mechanism. This is the refutation §8 promised to record. Three qualifications fix its scope:

1. **The candidate is itself open.** §8 applies "if the candidates hold". Chain Theory's 197.5 is an *open result* on its own side: the ladder is unmet at the sinks of its census. So the refutation is of the *identification of the two accounts*, conditional on Chain Theory's.
2. **L14's grade cannot fall.** L14 was a POSTULATE and remains one. What falls is §6.1's route to GENUINE ("if both hold"). §6.1's fallback ("if (a) holds and (b) fails, L14 becomes 'bounded above by theorem; selection open'") is **not** triggered. The bound of §A2 counts attach sites, and R2 shows those are not the Tree's generations; to adopt the bound, the Tree would have to adopt Chain Theory's account of generations in place of its own.
3. **The consequence for *The Forced and the Chosen*.** It staked the hardening of L14 from POSTULATE toward GENUINE. That hardening must now come from the quaternion side itself, not from Chain Theory's attach sites.

What the two threes share. One structure: an **oriented cyclic Z_3**.

- In $\mathbb{H}$, the cycle $i \mapsto j \mapsto k \mapsto i$ is an inner automorphism (`cycle_is_inner` , conjugation by $u = (1 + i + j + k)/2$). No automorphism swaps two units while fixing the third

(`no_orientation_reversing_swap`): ijk = −1 while kji = +1, so the quaternion triple has symmetry Z_3, not S_3.

- In Chain Theory, the triangle is carried by a directed shift.
- The script `q5_two_threes.py` shows the cycle, written on Im $\mathbb{H}$ in the basis (i, j, k), *is* the site shift. Its eigenbasis against (i, j, k) is the order-three Fourier matrix, every squared modulus 1/3 and every phase a multiple of a third of a turn, which is Chain Theory's eq. (250).

This sharing is real but generic: any triple carried by a cycle has it. We do not count it as evidence. Chain Theory's polar unitary of the incidence matrix (197.4) is a rotation by 40.22°, not of order four and not about the cycle's axis. Like every rotation, it is an automorphism of the quaternion triple, so it carries no discriminating content.

A4. R3 · the read cap lands on the Tree's colour block

The Tree's object. $M_{16} = \mathrm{End}(M_4) = M_4 \otimes M_4\text{^op}$ acts on the carrier $M_4 = \mathbb{C}^4 \otimes \mathbb{C}^{4*}$. The cut between the two factors is the bimodule's left–right cut (§A5). L13 splits $\mathbb{C}^4 = \mathbb{C}^3 \oplus \mathbb{C}^1$, so the carrier has a **gluon corner** $\mathbb{C}^3 \otimes \mathbb{C}^{3*}$, the $3 \otimes \bar{3} = 8 + 1$ of colour.

Shape (ii): the threshold reappears in the bimodule. Confirmed. Horodecki's 3×3 state (parameter $a = 0.3$) was placed on the gluon corner (`q6_read_cap_on_colour_block.py`) and checked:

- positive, with trace 1;
- a positive partial transpose across the left–right cut;
- entangled (realignment norm $1.003 > 1$).

It is therefore bound entangled, and no distillation can sharpen it. Against Chain Theory's read cap $d_1 d_2 \le 6$, the Tree's blocks sort as follows:

- **readable:** the isospin pair (2,2), the quark doublet (3,2) (exactly at the cap) and colour–lepton (3,1);
- **dark:** the gluon corner (3,3).

The cap falls exactly between the quark doublet and the gluon corner, and L23 is about the gluon corner.

Shape (i): a statement about the Dirac operator. Refuted. The statement was *"no conditional expectation onto a mutually-unbiased-basis subalgebra of the colour factor commutes with D on that block."* It is false for every colour-blind $D = 1_3 \otimes Y$, which is the form of every finite Standard-Model Dirac operator. Such a D commutes with the whole colour factor, and so with every conditional expectation onto a colour subalgebra; this was checked in the computational and the Fourier bases.

Against §6.2. §6.2 said that if (a) held, L23 would move to "mechanism stated, continuum limit open". We do not take that step. What holds is a *location*: Chain Theory's dark object exists on the Tree's carrier, at the colour block, across the cut where the Tree's left and right actions meet. It is not a mechanism for a mass gap, because no finite statement about (A, H, D) implies a gap. There is also a reason that applies to both programmes. Pure SU(3) Yang–Mills has a deconfining transition at finite temperature, with the same colour dimension on either side of it, so a dimension count alone cannot be what confines. A mechanism must enter through the state or the dynamics. **L23 stays ◇ OPEN**, and R4 holds: G1, G3 and G4 are untouched.

A5. R9′ · the improper element at M16, and L12 re-posed

v1's §7.1 found that R9 was posed on the wrong object, and re-posed it at M16 = End(M4) ≅ M4 ⊗ M4^op. There the Tree's L12 lives: the left factor $\mathcal{L}$ acts covariantly (ξ ↦ aξ) and the right factor $\mathcal{R}$ contravariantly (ξ ↦ ξb), as in `F2_3_ChiralityForced`.

The test. We used the Tree's γ1…γ4 (squares +1, +1, −1, −1), γ5 = γ1γ2γ3γ4 and P = γ1, and realised every canonical lift as a 16 × 16 matrix (`r9prime_parity_at_M16.py`). Each was tested on two things:

- the two factors;
- the carrier's split into 8 even and 8 odd Clifford elements, the eigenspaces of the grading Ad γ5.

Lift	the factors $\mathcal{L}$, $\mathcal{R}$	the 8 + 8 halves
L(P): ξ ↦ Pξ (the improper element acting on spinors)	preserved; acts on $\mathcal{L}$ alone	**exchanged**
R(P): ξ ↦ ξP	preserved; acts on $\mathcal{R}$ alone	**exchanged**
Ad P, the reflection, Ad γ5 (automorphisms)	preserved; act on both	preserved
reversion, Clifford conjugation, transpose (anti-automorphisms)	**exchanged**	preserved
J: ξ ↦ ξ† (antilinear; the real structure of a spectral triple)	**exchanged**	preserved

L(P) anticommutes with L(γ5). Ad P commutes with Ad γ5: the automorphism lift loses the sign that makes P a parity.

Outcome.

- R9′'s shape (ii) is realised. The improper element *singles out* one factor, and which one is the side the spinors are taken on, a convention.
- The kill-condition ("no action") is not met.

The new fact is that the two binary choices are carried by **different classes of operations**:

- exchanging the *factors*, which is L12 as the capstone words it ("which factor is left"), is done only by anti-automorphisms, of the kind that act as charge conjugation in noncommutative geometry;
- exchanging the *chiral halves* is done only by the improper element acting on spinors.

L12 is re-posed as two choices:

1. which factor is covariant: the 4 against the $\bar{4}$ of $(4,2,1) \oplus (\bar{4},1,2)$, a choice of the charge conjugation kind;
2. which chiral half SU(2)_L acts on: the parity choice.

The grade is unchanged (POSTULATE). The capstone's single sentence named the first choice; parity violation is the second.

A6. R11 · registers, and what Conjecture 0 does for L11

`q10_registers_vs_cascade.py` gives three results.

- **Every register Chain Theory uses is generated by the Tree.** Register 2 is the seed's carrier, 4 is the next level's, and 3 is L13's split, $4 = 3 \oplus 1$. The cascade's levels 16 and 256 are composites in Chain Theory, not registers.
- **The Tree's walked branch never needs a single system larger than 4.** The Pati–Salam factors are (4,2,2); after L13 they are (3,1,2,2); the fundamentals of su(4), su(3) and su(2) are 4, 3 and 2. This is consistent with R11 from the Tree's side. The census of Chain Theory's own move table remains Chain Theory's to report: its Conjecture 0 is a conditional theorem under that table.
- **By-product: L11 narrows.** L11's residue is that nothing in the mathematics prefers (4,2,2) among the three-factor decompositions of 16: (16,1,1), (8,2,1), (4,4,1) and (4,2,2). If every nontrivial factor must be one of Chain Theory's registers, only **(4,4,1) and (4,2,2)** remain, and the walked branch is one of them. The two survivors differ only in reading the second 4 as one ququart or as a qubit pair, which is the identification Chain Theory's Theorem 65.2 makes ("the hub pair as one qudit of dimension four"). This narrowing is conditional on Conjecture 0, like R7's narrowing of L9.

The six genres on the Tree's objects:

- (2,2), spacetime: agree (Q2: Chain Theory's β is the Tree's γ_1).
- (3,2): agree on the algebra. su(3)+su(2)+u(1), dimension 12, is both the cell's identification algebra and the algebra acting on the Tree's quark doublet.
- (4,2), weak: agree. It is the Pati–Salam (4,2,1) block, where SU(2)_L acts.

- (3,3), strong: agree. It is the gluon corner of §A4.
- The two Higgs cells, (4,3) and (4,4): open. The Tree's Higgs is the (1,2,2) bidoublet, $2 \otimes 2 = 4$ dimensions. Chain Theory's is a qubit pair read as one ququart (200.1, 65.2). They may be one object, met through 65.2. This is posed in §A8, not claimed.

A7. A correction of our own: the Higgs window and the measured Higgs mass

This test was not on v1's list. We ran it while comparing Chain Theory's Higgs quartic with the Tree's coupling window, and it found an error in how the Tree's window had been described.

The two numbers.

- **Chain Theory's quartic** (eq. 124 and Theorem 202.1) is the hub-face office over the vacuum office squared: $\lambda = (46656001681/182250000)/V^2$, with V at six readings (6561, 81, 65536, 256, 16, 256). At the first reading, $\lambda = 46656001681/7845264902250000 \approx 5.947 \times 10^{-6}$. Its convention, $m_H^2 = 2\lambda v^2$, matches the Tree's.
- **The Tree's window** (L15, `HiggsBridge`) is $g^2/N \le \tilde{\lambda} \le g^2$ at the cutoff Λ, with $N = 96$ and $\lambda = 4\tilde{\lambda}$. That is, $g^2/24 \le \lambda(\Lambda) \le 4g^2$, with $g = g_3(\Lambda)$.

Chain Theory states no scale for its number, so the comparison is conditional on reading that number at the Tree's cutoff.

The running. `q1_quartic_vs_window.py` evolves the Standard-Model couplings in the MS-bar scheme, with one loop in full plus the two-loop gauge terms and the dominant two-loop top, QCD and Higgs terms. The boundary values at m_t come from Buttazzo et al. (2013). **Validation:** the script gives $\lambda(M_{Pl}) = -0.0141$ against their -0.0143, and λ turns negative at $10^{9.7}$ GeV against their $\sim 10^{10}$.

Test	Result
Chain Theory's number at the cutoff ($\Lambda = 10^{13}$ … M_{Pl})	first reading: **below** the window at every cutoff, by a factor of 1,700–2,400. The reading V = 81 ($\lambda \approx 0.039$) is **inside** at every cutoff. V = 16 ($\lambda \approx 1.000$) is inside up to 10^{17} GeV.
Chain Theory's number as a running coupling	first reading: equal to the Standard Model's $\lambda(\mu)$ at $\mu \approx 10^{10.6}$ GeV, just below the scale where the measured λ crosses zero. This is a weak test, since any value in the Standard Model's range is met at some scale.
The Tree's window as a boundary condition at $\Lambda = 10^{16}$ … M_{Pl}	the lower edge needs $\lambda(m_t) \approx 0.139$, so **$m_H \approx 131$–$132$ GeV**; the upper edge needs $m_H \approx 168$–176 GeV. **The measured 125.20 GeV lies below the window.**
Robustness	the lower edge is strictly positive for every N ($\tilde{\lambda} = g^2 b/a^2 > 0$), so no choice of N helps. $\lambda(\Lambda) > 0$ with $m_H = 125.20$ needs $m_t < 171.0$ GeV (published NNLO: about 171.5), against the measured 172.57 ± 0.29.

What this corrects. The capstone makes no 125 GeV claim, and the window is a theorem about a defined quantity, so **L15's grade does not change** (PARTIAL). But the estate's `HiggsBridge` documentation said that running from Λ to the electroweak scale *produces* about 125 GeV, citing the literature. With Standard-Model field content it does not.

Read as a physical boundary condition at a high cutoff, the Tree's window **is in tension with the measured Higgs mass**: about 6 GeV in m_H, or about 1.5 GeV in m_t. This is the known difficulty of the spectral Standard Model, whose 2007 version predicted about 170 GeV. The literature reaches 125 GeV by adding a scalar singlet σ (Chamseddine–Connes 2012), and that singlet appears naturally in Pati–Salam spectral models (Chamseddine–Connes–van Suijlekom 2013). The Tree does not yet contain it.

The correction has been made where the claim appeared:

- a dated note in `HiggsBridge`, beside the original text, which is kept;
- the reference table `docs/WHERE_TODAYS_PHYSICS_SITS.md`;
- the Tree page at infinitography.com, which had also wrongly called 125 GeV "the Chamseddine–Connes value".

This correction was found by testing a Tree link against measurement, and the repair it calls for is now a staked prediction (R13).

A8. What has become sharper: the work before v3

Each item says what narrowed, what to compute, and what would confirm or refute. R13–R16 are new pre-registrations in the format of §7. The rest carry v1's residuals forward.

#	What	Narrowed from	Test	Confirms	Refutes (at that site)
R13	**The σ field.** The Tree's Pati–Salam finite geometry, with a Majorana entry in its finite Dirac operator, yields a scalar singlet σ, and with σ the boundary condition reproduces m_H ≈ 125 GeV	§A7	(a) the Tree's Pati–Salam algebra admits the Majorana entry and the inner fluctuation that produces σ (first-order condition relaxed, as in CCvS 2013); (b) the Standard Model plus σ, run from Λ with the Chamseddine–Connes 2012 boundary conditions	σ arises and m_H lands at 125.2 within the running's error	no Majorana entry or no singlet on the Tree's algebra: **L15 is then refuted as physics** at the Higgs mass, not merely in tension

#	What	Narrowed from	Test	Confirms	Refutes (at that site)
R14	**The Higgs cells.** Chain Theory's closed hub read (200.1), realised on $M_2 \otimes M_2$ (65.2), transforms as the Tree's (1,2,2) bidoublet under su(2) ⊕ su(2)	§A6	build the hub read on the two qubits and decompose it under the two su(2)s	(2,2)	any other representation: the two Higgs fields are different objects
R15	**Where the three generations live.** The Tree's own fermion count ($96 = 3 \times 16 \times 2$, the N = 96 of L15) carries the generations as a three-dimensional multiplicity space on which mixing acts. Decide which structure that space carries: the quaternion reading (real; symmetry SO(3); no intrinsic phase) or a qutrit reading (complex; Z_3 with ω; Fourier mixing)	§A3	(a) state L14 as a structure on the multiplicity space, not on $M_2(\mathbb{C})$; (b) compute which Yukawa textures respect each structure and whether the observed CP phase is compatible	the quaternion reading admits the observed mixing with its phase from a stated source	the quaternion structure forbids CP violation in quark mixing with no other source: **L14's quaternion reading is refuted**
R16	**The read cap through the vacuum.** If the read cap bears on confinement, it enters through the state: the vacuum of a minimal lattice gauge model, restricted to the colour block across the left–right cut, is bound entangled at strong coupling and not at weak coupling	§A4	the reduced vacuum state of a small SU(3) or Z_3 lattice gauge model, partially transposed, across a range of couplings	bound entanglement tracks the confining regime	no bound entanglement in the confining regime: the read cap is not the mechanism on the Tree's side
R1′	Formalise the triangle	§A2	state Chain Theory's closed core triangle as a structure in Mathlib and derive C′'s hypothesis from it	the count of three without network input	—
R10	The fourth level (from v1)	—	per-web dimension count of Chain Theory's sixteen particle webs, under the copy model, against M_{256}	all fit	a web needs M_{65536}
R6	The record space over M_{256} (from v1)	—	construct "one copy per link" from the inner-product lineage	the 172-record space	only the operator algebra (rank 66)
R5, R8, R12	The Born circle, the three aspects, Weinberg from the offices (from v1)	—	Q8, Q9, and the trace identification	as in §7	as in §7

#	What	Narrowed from	Test	Confirms	Refutes (at that site)
L11	Choose between (4,4,1) and (4,2,2)	§A6	whether Chain Theory's move table forces 65.2's qubit-pair reading	(4,2,2) forced	both admissible: L11's residue is one reading choice

R13 and R15 matter most. Each can refute a Tree link *as physics*, R13 at the Higgs mass and R15 at the quaternion account of generations, and each has a concrete computation behind it. We commit to running them before v3, and to recording either outcome here.

A9. Caveats, and provenance of this version

Caveats. Everything in §§A2–A6 is finite mathematics about the two programmes' objects; none of it verifies a physical identification on either side.

- §A7's running is one loop in full plus the dominant two-loop terms. It is validated against the published NNLO results to about 0.0002 in λ(M_Pl), and its conclusion (a few GeV) is far larger than that error.
- Chain Theory's quartic is quoted at its six readings, and the choice among them is Chain Theory's to make.
- Each refutation and correction above is conditional exactly as stated.

Provenance. The v1 text is unchanged above this addendum. The Lean files, scripts and the write-up were committed to `wonderben-code/convergence-codex` on 28 October 2026 (commits 13ad865 and 97c65cb), and every push to that repository is anchored to the Bitcoin blockchain by OpenTimestamps. This version's text is committed before it is deposited. The dated record establishes the priority of the *predictions*. It makes no claim on anyone's solutions, and every result of Chain Theory cited here is his.

References added in v2

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- A. H. Chamseddine and A. Connes, *Resilience of the spectral standard model*, JHEP 09 (2012) 104
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- P. Horodecki, *Separability criterion and inseparable mixed states with positive partial transposition*, Phys. Lett. A 232, 333 (1997)

- Particle Data Group, *Review of Particle Physics* (2024): m_t = 172.57 ± 0.29 GeV, m_H = 125.20 ± 0.11 GeV
- M. E. Mala, *Chain Theory ↔ the Tree of Reality: results, second pass*, `docs/chain-theory/09_results_second_pass.md` in `wonderben-code/convergence-codex` (28 Oct 2026)